EXPERIMENT 14

Sentence Agreement Checking

Aim

To check simple subject-verb agreement using rules based on a small context-free grammar.

Procedure

1. Define allowed subject and verb forms.
2. Split the sentence into words.
3. Check whether the subject and verb agree in number.
4. Display the agreement result.
5. Test singular and plural examples.

Program

```
def check_agreement(sentence):
    words = sentence.lower().replace(".", "").split()
    if len(words) != 2:
        return "Unsupported sentence pattern"
    subject, verb = words
    singular_subjects = {"he", "she", "it", "cat"}
    plural_subjects = {"they", "we", "dogs"}
    singular_verbs = {"runs", "sleeps"}
    plural_verbs = {"run", "sleep"}

    if subject in singular_subjects and verb in singular_verbs:
        return "Agreement correct"
    if subject in plural_subjects and verb in plural_verbs:
        return "Agreement correct"
    return "Agreement incorrect"

for s in ["he runs", "they run", "he run"]:
    print(s, ":", check_agreement(s))
```

Output

```
OUTPUT — EXPERIMENT 14
he runs : Agreement correct
they run : Agreement correct
he run : Agreement incorrect
```

Result

Thus, the program checked subject-verb agreement for the defined sentence patterns.